

Adrian Dar T. Serapio

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Research Interests

My research is centered on *Embodied Clinical Intelligence* with the goals of 1) building multimodal AI systems grounded in clinical context and diagnostic explainability, 2) architecting agentic systems that can support clinical workflows and decision-making, and 3) establishing the clinician-AI interface for seamless interaction and synergistic collaboration.

Education

UC Berkeley-UCSF, Berkeley, CA & San Francisco, CA 2025–Present
Ph.D. in Bioengineering, Designated Emphasis in Computational Precision Health
Advisors: Yang Yang, Kang Wang, and Jae Ho Sohn

UC Berkeley, Berkeley, CA 2020–2024
B.S. in Electrical Engineering and Computer Sciences (GPA: 3.78/4.00)

Research Experience

Graduate Researcher, Yang Lab, Sohn Lab — UCSF 2025–Present

- Lead the development of EHRGym, a reinforcement learning environment and benchmark for training and evaluating autonomous computer-use agents within clinical workflows.
- Lead the prospective validation of an EHR-grounded chest X-ray interpretation AI system using vision-language models to disambiguate nonspecific imaging findings.
- Lead the development of a multi-agent system for abdominal cancer cohort database curation, enabling downstream applications such as early pancreatic cancer detection and hepatocellular carcinoma lesion tracking.

Staff Research Associate, Sohn Lab — UCSF 2024–2025

- Led a multi-reader clinical reader study leveraging large language models to improve the clinical utility of referring clinician-provided indications.
- Mentored undergraduate researchers in building a patient-facing vision-language agent for lung cancer screening and developing an automated evaluation metric for patient-centered radiology report communication.

Undergraduate Researcher, Sohn Lab — UCSF 2021–2024

- Led the development of an open-source fine-tuned large language model designed to generate radiology report impressions from the radiology report findings and a multi-reader study to assess model performance against radiologists.

Undergraduate Researcher, Data-Intensive Development Lab — UC Berkeley 2020–2021

- Designed geospatial and statistical visualizations (choropleth maps, parity plots) of machine learning-based wealth estimates in Togo, which were featured in a *Nature* cover article.

Work Experience

Software Development Engineering Intern, Amazon Summer 2023

- Developed an observability pipeline for Ring Data Management to attribute infrastructure costs to internal services by aggregating time-series logs into daily and monthly cost estimates.

Selected Publications

- **Serapio, A.***, Chen, T.L.*, Tangsombatvisit, B., Fields, B.K.K., Yu, Y., Guo, Y., Kim, S.K., Miao, B.Y., Sushil, M., Hess, C.P., Majumdar, S., & Sohn, J.H. Radiologically Relevant Clinical History Summarization with Large Language Models: A Multi-Reader Performance Study. *Radiology*, Accepted (In Press).
- **Serapio, A.**, Chaudhari, G., Savage, C.H., Lee, Y.J., Vella, M., Sridhar, S., Schroeder, J.L., Liu, J., Yala, A., & Sohn, J.H. An Open-Source Fine-Tuned Large Language Model for Radiological Impression Generation: A Multi-Reader Performance Study. *BMC Medical Imaging*, 24:254, 2024.

Technical Skills

Programming: Python, JavaScript, R, Bash

Frameworks & Tools: PyTorch, Hugging Face, NumPy, Pandas, Matplotlib, Node.js, Next.js, Git, AWS, HPC/Slurm